

BIOLOGY GRADE 11: EXCRETION IN PLANTS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.3 (5 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 2.0: Anatomy and Physiology of Plants
Sub-Strand	Sub-Strand 2.3: Excretion in Plants
Total Duration	5 lessons × 40 minutes = 200 minutes total
Sub-Strand Content	– Meaning and importance of excretion (include reasons why plants lack complex excretory structures) – Methods of excretion in plants (transpiration, diffusion, deposition through organ/tissue fall, exudation and guttation) – Economic importance of plant excretory products
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe ways through which plants eliminate waste products, - demonstrate ways through which plants eliminate waste products, - outline the economic importance of plant excretory products, - recognise the importance of excretory products in plants.
Core Competencies	– Learning to Learn: The learner shares information with others discussing plant excretory products and their economic importance. – Self-efficacy: The learner enhances self-confidence while carrying out experiments to demonstrate ways through which plants eliminate waste products.
Core Values	– Respect: The learner is considerate of others' opinions while brainstorming on reasons why plants lack complex excretory structures. – Integrity: The learner demonstrates honesty while reporting findings on ways through which plants eliminate waste products.
Science & Engineering Practices	– Asking Questions and Defining Problems: Students ask questions about how plants get rid of wastes without kidneys, and add them to the Driving Question Board. – Developing and Using Models: Students build and revise a cumulative model of plant waste production, recycling, storage and elimination across all five lessons. – Planning and Carrying Out Investigations: Students carry out a school nature walk, a bell jar guttation setup and leaf microscopy using Acacia, Euphorbia and potted seedlings. – Analyzing and Interpreting Data: Students compare fresh and aging leaves and categorize plant excretory products by source and use. – Engaging in Argument from Evidence: Students debate why plants do not need kidneys and use a Claim-Evidence-Reasoning (CER) matrix to synthesize community interview findings. – Obtaining, Evaluating and Communicating Information: Students research plant products such as quinine, rubber, gums and

	tannins and present a portfolio.
Pertinent & Contemporary Issues (PCIs)	– Drug and Substance Abuse: The learner is sensitised about the effects of substance abuse while conducting interviews with members of the local communities about the uses of various plant products. – Financial Literacy: The learner appreciates the commercial value of plant excretory products while exploring and brainstorming on their economic importance.
Career Connections	– Pharmacologist: studies how plant alkaloids such as quinine and morphine act on the body. – Industrial Chemist: extracts and processes latex, tannins, gums and essential oils. – Agricultural Officer: advises farmers on growing and harvesting commercial crops such as tea, coffee and rubber. – Botanist: studies plant structure, physiology and waste handling. – Pharmacognosist: identifies medicinal compounds from plants. – Forestry Officer: manages sustainable harvesting of gums and resins from trees such as Acacia. – Commercial Agro-processor: turns plant products such as tea, coffee and papain into marketable goods.
Focus for Lessons	This unit uses puzzling sights from the school neighbourhood, such as resin oozing from cut Acacia and Euphorbia, guttation droplets on leaf edges and seasonal leaf fall, to help learners explain how plants remove wastes without kidneys. Through predictions, nature walks, experiments, microscopy, research and community interviews, learners build and revise a model of plant excretion and its economic value, and consider the health risks of some alkaloids and the need for sustainable harvesting.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets? KICD KEY INQUIRY QUESTIONS: 1. How do plants eliminate waste products? 2. How do humans use plant excretory products?
Anchoring Phenomenon	When gum arabic (Acacia) or Euphorbia trees in the school neighbourhood are cut, thick sticky milky sap or amber resin oozes out and hardens into a sealant. On early humid mornings, leaves of plants in the garden, such as grass and arrowroot (taro) leaves, have droplets of liquid standing along their edges, while old yellowing leaves drop off seasonally. This is puzzling because plants have no kidneys, bladders or sweating systems, so students wonder how plants get rid of their wastes, whether these oozing liquids and falling leaves are really waste removal, and why some of these products are valuable in Kenyan markets.
Storyline Thread	Lesson 1: Excretion is removing metabolic wastes; plants lack complex excretory organs because their metabolic rates are low, they recycle wastes (CO₂ and O₂), and store insoluble wastes in tissues. Students compare human and plant metabolic rates and wastes through brainstorming, analyze CO₂/O₂ recycling in photosynthesis and respiration, and debate why plants do not need kidneys, using a chart of plant vs. animal metabolic rates and diagrams of carbon and oxygen recycling in mesophyll cells. Leads to: If plants do not have organs to flush out wastes, how do they actually get rid of the wastes they produce? Lesson 2: Plants eliminate gaseous and liquid wastes through diffusion (stomata/lenticels), transpiration (water vapor), exudation (gums/resins), and guttation (hydathodes under high humidity). Students conduct a nature walk around the school locality to observe exudation (resin/latex) and tissue drop, and set up a potted seedling experiment under a bell jar to observe guttation droplets at hydathodes, using local trees (Acacia, Euphorbia), potted plants, bell jar, plastic bags, hand lenses and an observation logbook. Leads to: What happens to non-gaseous or insoluble waste products that cannot evaporate or ooze out easily? Lesson 3: Insoluble metabolic wastes are deposited in non-vital organs (bark, old leaves, fruits, flowers) which are periodically shed through tissue fall. Students dissect and compare young fresh green leaves with aging yellow/brown leaves from school trees and test tissues for insoluble calcium oxalate crystals under a light microscope, using razor blades, slides and iodine/dilute acid

	solutions. Leads to: Since plants produce all these chemical excretory compounds, can humans put them to economic or industrial use? Lesson 4: Plant excretory products (latex, tannins, alkaloids like quinine and caffeine, gums, essential oils) have widespread economic, medicinal, and commercial value. Students search reference materials and digital resources to categorize plant excretory products by type, plant source and economic application, and discuss their commercial value in Kenya (quinine, rubber, papain, gums, tannins). Leads to: How do local community members make practical use of these plant excretory products, and are any harmful? Lesson 5: Local communities utilize plant excretory products for medicine, sealing, and tanning, but some alkaloids pose health risks (drug and substance abuse). Students conduct community interviews and peer project research on uses of local plant products using survey guides and interview notes, present a portfolio on poster paper, and synthesize findings using a Claim-Evidence-Reasoning (CER) matrix. Leads to: How can sustainable harvesting of useful plant excretory products be balanced with environmental conservation?
Supporting Phenomena	– Resin and gum collected from Acacia trees in Rift Valley and drylands is sold as gum arabic, even though it is a plant product that oozes out of the tree. – Latex tapped from rubber trees and Euphorbia hardens into sticky material that communities use for sealing. – Tea in Kericho and coffee in Central Kenya contain caffeine, an alkaloid, yet these crops are grown for sale at home and abroad. – Old yellow leaves and bark shed from trees on the school compound carry stored wastes away from the plant. – Quinine from the cinchona tree is used to treat malaria, while other plant alkaloids such as morphine can be harmful if abused.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.

LESSON 1 (80 minutes (double lesson: 2 x 40 min)): Where Do Plant Wastes Go? Gum, Droplets and Falling Leaves

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon (oozing sap and resin from Acacia and Euphorbia, droplets on grass and arrowroot leaf edges, and seasonal leaf fall) and to help learners build an initial explanation of what excretion is and why plants lack kidneys or other complex excretory organs.
Knowledge	Learners discuss the meaning and importance of excretion, and give reasons why plants lack complex excretory structures: (1) low metabolic rates compared with animals, (2) recycling of wastes such as CO ₂ and O ₂ between photosynthesis and respiration, and (3) storage of insoluble wastes in tissues.
Skills	Developing and using models (comparative diagrams of human and plant waste removal); asking questions (sticky-note questions for the DQB); analysing and interpreting data (chart of plant vs animal metabolic rates); engaging in argument from evidence (debate on why plants do not need kidneys); collaboration in groups of 5.
Attitudes	Respect: the learner is considerate of others' opinions while brainstorming on reasons why plants lack complex excretory structures. Learning to Learn: the learner shares information with peers. Curiosity about local plants and their economic value (Financial Literacy: the learner appreciates the commercial value of plant excretory products).
Key Inquiry Question	1. How do plants eliminate waste products? 2. How do humans use plant excretory products?
Purpose in Storyline	This opening lesson establishes that excretion is the removal of metabolic wastes and that plants manage without kidneys for specific reasons. It surfaces learners' initial ideas, opens the Driving Question Board and ends with the puzzle that drives Lesson 2: if plants do not have organs to flush out wastes, how do they actually get rid of the wastes they produce?
Safety Notes	Euphorbia latex is an irritant and can cause skin burns and serious eye damage: the teacher alone handles or cuts it, and learners observe from at least 1 m and NEVER touch it or rub their eyes. Do not taste any sap, resin or gum. Learners wear closed shoes on the nature walk, stay in their groups, avoid thorny Acacia branches, and wash hands with soap after the walk. Check for allergies before the walk.

B. LESSON OVERVIEW

Learners walk the school compound and meet the phenomenon first-hand: a cut Acacia/Euphorbia oozing milky sap or amber resin, droplets along the edges of grass and arrowroot (taro) leaves on a humid morning, and yellowing leaves dropped under trees. They sketch a comparative diagram of how a human (kidneys, skin, lungs) and a plant get rid of wastes, and post questions on the DQB. They then compare human and plant metabolic rates using a chart, analyse diagrams of CO₂ and O₂ recycling in mesophyll cells, and debate 'Plants do not need kidneys'. The lesson closes with a Claim-Evidence-Reasoning (CER) statement, an exit ticket and the question that launches Lesson 2.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Minutes 1-25. Learners go on a short Nature Walk (about 10	Before the lesson, at dawn or early morning, check the	Predict-Observe-Explain (prediction step) combined with	Teacher observation during the walk: tick on a class list who points

VIDEO: Cellular respiration Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: excretion plants metabolic wastes	minutes) around the school compound in their groups of 5 with hand lenses and notebooks. They observe a cut Acacia branch with gum arabic (or a fresh sample brought by the teacher), a Euphorbia candelabra stem with milky latex (seen from a safe distance), droplets on the edges of grass and arrowroot leaves, and fallen yellow leaves under a tree. Back in class, each learner draws a comparative diagram: on the left, a human showing where wastes leave (kidneys to urine, skin to sweat, lungs to CO₂); on the right, a plant showing where they THINK its wastes go, with labels and arrows. They write one prediction: 'I think a plant gets rid of its wastes by...' Learners then Think-Pair-Share their drawings with a partner.	arrowroot/grass droplets exist and that a safe Acacia/Euphorbia sample is ready. At the start say: 'Today we are detectives. A tree in our compound is leaking something. Is it sick, or is it getting rid of something?' Outside, set rules: 'Look, do not touch. I will handle the Euphorbia.' Demonstrate the cut yourself and keep learners 1 m back. Ask: 'What do you notice? What do you wonder?' Give WAIT TIME of 10-15 seconds before taking answers. Back inside, say: 'Draw what you honestly think, not what you think I want. Wrong ideas are welcome today.' During Think-Pair-Share, give 2 minutes thinking, 3 minutes pairing, then cold-call 6 pairs (not volunteers) to share one idea. Record the ideas on the board without judging: 'Thank you, I am writing that down exactly as you said it.' Circulate and note common ideas such as 'plants sweat', 'wastes stay in the roots', 'the sap is blood'.	Nature Walk and Direct Field Observation. Making a drawn prediction before any teaching commits learners to an idea they can later test and revise. The walk anchors the whole unit in real Kenyan plants, so later explanations always return to the oozing sap, leaf-edge droplets and leaf fall.	out at least 2 distinct features (gum or latex, droplets, fallen leaves) and who stays at a safe distance. Check drawings for observable indicators: (a) human diagram shows at least 3 routes (kidneys, skin, lungs), (b) plant diagram has at least 1 labelled exit route or states 'don't know'. Tally on the board the number of learners who think plants have an organ like a kidney, sweat, or keep wastes inside, to use as a baseline.
Observe Phase VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: excretion plants	Minutes 26-50. Learners gather evidence in three steps. (1) Brainstorm in groups of 5: 'What are metabolic wastes? Name those humans produce (CO₂, urea, water, excess salts) and those plants might produce.' (2) Each group receives a printed Chart comparing plant vs. animal metabolic rates (e.g., a sprinting Kenyan athlete or a running goat/zebra compared with a maize plant or Acacia tree: oxygen use, food use per kg of body mass, and amount of nitrogenous waste made, with plants much lower). They annotate: 'What does the chart show? What does it suggest about the amount of waste plants	Distribute charts and diagrams (one per group of 5, 8-10 sets for the class). Say: 'You have 3 minutes of silent thinking, then talk.' For the brainstorm, give WAIT TIME of 10-15 seconds after each question, then cold-call 5 groups for different examples of wastes and record them in two columns (human, plant). Ask: 'A runner at the Iten training camp breathes hard and sweats. Does a maize plant in Kitale do that? Why or why not?' Prompt with: 'What does the chart tell you about how much waste a plant makes compared with a person?' For the diagrams, ask: 'Where does the CO₂ from respiration go in	Brainstorming and data analysis with a chart and diagrams. Comparing quantitative patterns (low plant metabolic rate) and a cycle diagram (recycling) lets learners move from 'plants must have hidden kidneys' to evidence-based reasons, while the word excretion is introduced only after learners have a concrete sense of metabolic waste.	Observable indicators: (a) each group lists at least 3 metabolic wastes correctly and separates excretion from egestion; (b) learners state in their own words that plants make less waste per unit mass because their metabolic rate is lower; (c) learners correctly trace CO₂ and O₂ arrows in the mesophyll diagram. Teacher checks the annotated chart and notes groups that confuse photosynthesis with respiration, and re-teaches by asking 'Which process uses light? Which one releases energy?' at that group's table.

metabolic wastes	make?' (3) Groups study diagrams of carbon and oxygen recycling in mesophyll cells: CO ₂ from respiration used in photosynthesis, and O ₂ from photosynthesis used in respiration. They trace the arrows with a finger and a coloured pencil and answer: 'Which wastes of one process are raw materials of the other?' Groups also note that insoluble wastes such as resin, latex, gum and tannins are stored in vacuoles, bark, heartwood and old leaves.	daytime?' and 'Where does the O ₂ from photosynthesis go?' Wait 10-15 seconds, then cold-call 4 learners to explain the arrows. Probe: 'So is that really a waste, if the cell can use it again?' Introduce the term excretion: 'Excretion is the removal of metabolic wastes from the body.' Clarify that egestion (removing undigested ugali fibre) is not excretion. Circulate and ask each group one 'why' question. Ensure a respectful tone: 'We listen to every idea before we respond.'		
Explain Phase  VIDEO: Cellular respiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: excretion plants metabolic wastes	Minutes 51-68. The class holds a structured debate on the motion: 'Plants do not need kidneys.' Half of the groups argue FOR and half AGAINST (the teacher assigns sides, and sides switch after 4 minutes). Each group builds a Claim-Evidence-Reasoning (CER) statement on a strip of paper using the chart and diagrams. Example: Claim: Plants do not need kidneys. Evidence: plants have low metabolic rates and make little nitrogenous waste; CO ₂ and O ₂ are reused in the mesophyll; resin, latex and gum are stored in tissues. Reasoning: because little waste is produced, much is recycled, and the rest is stored as insoluble, harmless substances, no complex organ is needed. Learners also connect to the phenomenon: 'The Acacia gum and Euphorbia latex are stored wastes or by-products which ooze when the tissue is cut, and yellow leaves carry stored wastes away when they fall.' They state which parts of the phenomenon are still unexplained, such as the leaf-edge droplets.	Set debate rules: 'Use evidence from the chart or diagram. We challenge ideas, not people.' Give each side 3 minutes of preparation. Run the debate: allow 2 speakers per side in each round, call on non-volunteers (cold-call at least 8 learners overall), and enforce WAIT TIME of 10-15 seconds after each rebuttal question such as 'What evidence supports that?' After the switch, ask: 'Did arguing the other side change your thinking?' Provide the CER sentence frame on the board: 'I claim... because the evidence shows... which means...'. Then say: 'So if plants do not have kidneys, are they really free of waste problems?' Highlight gaps: 'We have explained why plants do not need kidneys, but we have not yet explained how the wastes actually leave or get stored. What about the droplets on the arrowroot leaves?' Praise the quality of reasoning, not the side that wins. Confirm key ideas: excretion definition, three reasons for lack of complex organs.	Structured debate with Claim-Evidence-Reasoning (CER). Arguing from evidence and switching sides forces learners to use the data, tests their reasoning and exposes the limits of their current model; the CER frame turns opinions into evidence-based explanations that are checked in later lessons.	Use a 3-point CER check on the strips: 1 point for a clear claim, 1 for at least 2 pieces of evidence from the chart or diagrams, 1 for reasoning that links evidence to claim. Observable indicators during debate: learners cite 'the chart shows...' rather than 'I feel...', and they listen respectfully (Respect value). Note learners who can name all three reasons. Exit ticket (last 2 minutes, on a half sheet): 'Give one reason why plants do not need kidneys, and name one plant product from our walk that may be a stored waste.'

Driving Question Board (DQB) Creation  VIDEO: Cellular respiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: excretion plants metabolic wastes	Minutes 69-76. The Driving Question is displayed: 'How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets?' Each learner writes 2 sticky-note questions about the phenomenon (for example: 'Why do arrowroot leaves have water drops in the morning?', 'Is gum arabic really a waste?', 'Why is Euphorbia latex dangerous?', 'Who buys gum arabic in Kenya, and for what?', 'Why do leaves turn yellow before falling?'). Learners post their drawings and notes on the DQB and, in groups, cluster questions under headings such as 'How wastes leave', 'What the wastes are', 'Uses and money value'. They link at least one question to the KICD inquiry questions.	Prepare a large chart paper or wall space labelled DQB with the Driving Question at the top. Say: 'Good scientists ask questions they do not yet know the answers to. Write two questions you really want answered.' Model one: 'Why do leaf edges hold droplets and not the middle?' Give WAIT TIME of 10-15 seconds for writing. Cold-call 5 learners to read a question aloud and ask the class: 'Where should this go?' Invite classmates to group them. Point to the unanswered ones: 'We cannot answer these yet. Which of them will our experiments help with?' Mention the economic angle: 'Gum arabic from Acacia is traded in Turkana, Marsabit and Garissa regions and Euphorbia hedges are common in Kisumu and Machakos. Why might someone pay for this?' Photograph the DQB at the end of the lesson to track later changes.	Driving Question Board with student-generated questions. Giving learners ownership of questions shows that the unit follows their curiosity, connects each lesson back to the phenomenon, and provides a visible record of how understanding grows.	Count and review sticky notes: each learner has 2 questions; at least half of them go beyond yes/no (start with How or Why). Observable indicators: questions mention the phenomenon (sap, droplets, leaf fall), and some link to money or uses. Use the first DQB check-in to sort questions into: answered today, to be answered in Lesson 2 onwards, and project questions about economic value.
Model Building Phase  VIDEO: Cellular respiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: excretion plants metabolic wastes	Minutes 77-80 plus home task. Learners return to their Day 1 comparative diagram and revise it with a different coloured pen: adding the three reasons plants lack kidneys (low metabolic rate, recycling of CO₂ and O₂, storage of insoluble wastes) and marking with a '?' the routes by which the wastes leave that they cannot yet explain. Each learner adds a short caption: 'Plants do not need kidneys because...' They write the new question for the next lesson: 'If plants do not have organs to flush out wastes, how do they actually get rid of the wastes they produce?' For the project introduced today, learners choose one local plant that produces gum, resin, latex or oil, and plan to	Say: 'Take out your first drawing. In a new colour, show what you now know, and mark with a question mark what you still cannot explain.' Give WAIT TIME of 10-15 seconds before asking: 'What did you add that was not in your first drawing?' and cold-call 3 learners. Name the gap: 'We know why there are no kidneys, but how do the wastes get out? That is our puzzle for next lesson, when we set up a guttation experiment using a Predict-Observe-Explain approach.' Introduce the portfolio project with a simple rubric (plant name, product, uses, photograph, name of the community member, approximate price in a Kenyan market) and mention peer evaluation of portfolios later. Give	Cumulative model revision. Revising the initial drawing with a second colour makes the growth in learners' thinking visible and prepares them to keep refining the same model across the unit; the leftover '?' marks turn gaps into purposeful questions for investigation.	Compare the revised and the original drawing: indicators of progress include (a) at least 2 of the 3 reasons written correctly on the model, (b) the term excretion used accurately, (c) a question mark placed on the unexplained exit routes. Review exit tickets and sort into: 3 reasons correct, 1-2 reasons, none. Use this to plan a quick 3-minute recap at the start of Lesson 2 for learners who are still unsure. Record who has chosen a project plant.

	interview a community member (a parent, elder, trader at a market or farmer) about its uses, taking a photograph of the plant and its product for the portfolio. They are reminded to stay safe and respect the interviewee's views, including that some plant products can be abused as drugs.	ethical interview guidance: 'Ask politely, get permission before taking photos, and report only what you were told (Integrity). If an interviewee mentions plants used as drugs, note the dangers of substance abuse.' Collect exit tickets at the door.		
--	---	--	--	--

D. TEACHER REFLECTION

After the lesson: Did learners see the sap, droplets and leaf fall as linked to waste removal, or as unrelated curiosities? Which prior ideas (plants 'sweat', 'sap is blood', 'wastes stay in the roots') were most common on the Day 1 drawings, and did they shift on the revised drawings? Was the chart of metabolic rates understood by all groups, including learners who struggle with data? Did the debate involve quieter learners through cold-calling? Was Euphorbia latex handled safely at all times? Were groups of 40-50 learners manageable on the nature walk, and do I need more adult help or smaller walking groups next time? Note which DQB questions to address in Lesson 2 (guttation and the leaf-edge droplets) and prepare materials (potted seedlings of maize or beans, transparent plastic covers, hand lenses) for the POE guttation setup.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed gum or resin and milky latex oozing from cut Acacia and Euphorbia, droplets along the edges of grass and arrowroot leaves in the early morning, and yellow leaves falling from trees. The chart showed plants have much lower metabolic rates than animals, and the mesophyll diagrams showed CO ₂ and O ₂ being passed between respiration and photosynthesis.
What did I learn?	Excretion is the removal of metabolic wastes from the body. Plants lack complex excretory organs like kidneys because (1) their metabolic rate is low so they make little waste, (2) they recycle CO ₂ and O ₂ between respiration and photosynthesis, and (3) they store insoluble wastes such as resin, latex and gum in tissues like bark, vacuoles and old leaves.
How does this explain the phenomenon?	Claim: Plants do not need kidneys. Evidence: low metabolic rates, recycling of CO ₂ and O ₂ in mesophyll cells, and stored products like gum arabic and latex that ooze out when tissue is cut. Reasoning: little waste is made and much of it is reused or stored harmlessly, so no complex organ is needed. Still unexplained: how the remaining wastes actually leave the plant, such as the leaf-edge droplets, and why some products are valuable in Kenyan markets.

LESSON 2 (2 double lessons (approximately 80-90 minutes each)): How Do Plants Get Rid of Wastes? Nature Walk, Exudation and Guttation Evidence

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather first-hand evidence of the ways plants eliminate wastes (diffusion, transpiration, exudation, guttation and tissue fall) through a nature walk and a bell jar guttation experiment, then use Claim-Evidence-Reasoning to explain how plants without kidneys still excrete.
Knowledge	By the end of the lesson, the learner should be able to: a) describe ways through which plants eliminate waste products (diffusion through stomata and lenticels, transpiration, exudation, guttation, organ/tissue fall); b) demonstrate ways through which plants eliminate waste products.
Skills	Planning and carrying out investigations (NGSS SEP 3); Analysing and interpreting data (SEP 4); Constructing explanations (SEP 6); Engaging in argument from evidence (SEP 7); careful observation and recording with a hand lens; reporting findings to peers.
Attitudes	Respect: considerate of others' opinions while brainstorming and debating. Integrity: honest reporting of what was actually observed, including when guttation droplets are few or absent. Self-efficacy: confidence while carrying out experiments. Learning to Learn: shares information with peers. Financial Literacy: begins to notice the commercial value of gum, resin and latex.
Key Inquiry Question	1. How do plants eliminate waste products? 2. How do humans use plant excretory products? (Introduced through observation of gum arabic and Euphorbia latex.)
Purpose in Storyline	Lesson 1 established what excretion is and why plants lack complex excretory organs. Lesson 2 provides the first direct evidence of HOW wastes leave plants: gaseous and liquid wastes leave by diffusion, transpiration, exudation and guttation. It answers part of the driving question and raises a new puzzle for Lesson 3: What happens to non-gaseous or insoluble wastes that cannot evaporate or ooze out easily?
Safety Notes	Euphorbia latex is toxic and an irritant: learners must NOT touch it, rub eyes, or taste it; observe from a distance with a hand lens, and only the teacher may make a tiny cut with a clean blade while wearing gloves or using a twig. Wash hands with soap and water after the walk. Stay in groups, within agreed boundaries, with wide-brimmed hats or water as needed. Beware thorns on Acacia: wear closed shoes and do not pull branches. Do not taste any resin or sap. Handle bell jars (glass) with two hands; if no bell jar, use a clear plastic container or bag. Report any skin contact immediately.

B. LESSON OVERVIEW

The lesson begins with a prediction about the school-locality phenomenon: why does milky sap ooze from a cut Euphorbia, and why are there droplets on arrowroot and grass leaves at dawn? Learners (40-50 in 8-10 groups of 5) predict, then go on a nature walk around the school grounds to observe exudation on Acacia and Euphorbia, leaf scars, fallen yellow leaves, and early-morning droplets on grass and arrowroot leaves. Back in class, groups set up a potted seedling under a bell jar or plastic bag (Predict-Observe-Explain) to generate guttation droplets in high humidity, with a control, and view leaf edges with hand lenses. Learners use Think-Pair-Share and Claim-Evidence-Reasoning to explain which wastes leave by which route, distinguishing guttation from transpiration. They add questions to the Driving Question Board (DQB) and revise their cumulative class model, ending with the new question about insoluble wastes. Note: guttation droplets overnight/early morning need to be checked the next day, so the experiment is set up at the end of Day 1 and read on Day 2.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Diffusion - Introduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Diffusion (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the Lesson 1 driving question and look at a photo or the teacher's sample of dried gum arabic and a picture of milky Euphorbia sap and dew-like droplets on arrowroot leaf edges. Individually, in their logbooks, they write: (1) Where do I think the sap and droplets come from? (2) Are they wastes, water, or something else? (3) What do I predict I will see on the nature walk and under the bell jar tonight? Then Think-Pair-Share with a neighbour, and groups draw a quick 'prediction sketch' of a plant with arrows showing where wastes might leave.	Say: 'Last lesson we agreed plants have no kidneys or bladders. Today you are detectives: how do they get rid of wastes anyway?' Display the gum arabic sample and ask: 'Where in Kenya might this be collected and sold?' Give 2 minutes silent writing, then WAIT TIME of 10-15 seconds after asking: 'What do you predict will come out of a cut Euphorbia stem, and is it waste?' Pair-share for 2 minutes, then cold-call 6 learners from different groups, recording predictions on the board in a 'Predictions' column without saying right or wrong. Ask: 'Who disagrees, and what is your reasoning?' Remind about respect for others' opinions and Euphorbia safety rules before leaving.	Predict-Observe-Explain (POE) and Think-Pair-Share: eliciting prior ideas makes learners' existing thinking visible, so they can later compare predictions with evidence and notice where their ideas change. Surfaces common misconceptions (e.g., plants 'sweat' like humans, all droplets are dew, leaf fall is only about seasons).	Observe whether prediction sketches name at least one exit route (leaves, stem, bark, roots). Indicators: learners distinguish 'dew' from 'plant liquid'; use the word 'waste' in predictions; give a reason, not only a guess. Tally misconceptions on the board (e.g., 'plants sweat', 'dew only') to revisit in the Explain phase.
Observe Phase  VIDEO: Diffusion and osmosis - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Diffusion Practice Source: CK-12  Search ARES for similar readings	Part A, Nature walk (about 40 minutes): In groups of 5, learners walk the school locality with a hand lens and logbook. Station tasks: (1) Observe a gum arabic Acacia trunk for hardened amber gum and resin on bark; (2) Observe a Euphorbia (candelabra tree or hedge) from a safe distance where the teacher makes a tiny cut and milky latex appears; record colour, texture, how fast it dries, and what happens to the sealed cut; (3) Examine lenticels on a young stem or bark with a hand lens; (4) Collect fallen yellow leaves, seed pods or bark flakes and note leaf scars as evidence of tissue/organ fall; (5) Look at grass and arrowroot (taro) leaf edges for droplets. Learners sketch, tabulate and photograph (if a phone is	Before the walk: 'Your job is to collect evidence, not to touch Euphorbia sap. Hands behind your back at that station.' Assign roles (recorder, lens-holder, safety monitor, sketcher, timekeeper). Circulate and ask each group: 'What exactly do you see? How is this different from water?' and 'What do you think this tells us about waste?' Use WAIT TIME of 10-15 seconds before prompting. Cold-call at least 2 learners per group during the walk to describe an observation. At the experiment set-up say: 'Write your prediction first: what will be on the leaf edges tomorrow and why?' Check that the seedling is well watered, jars are sealed against the soil surface or plastic, and the control is labelled. Remind learners that	Nature Walk and Direct Field Observation plus POE: real trees in the learners' own locality make the phenomenon concrete, and structured observation tasks focus attention on patterns (gum hardens, latex seals wounds, droplets at the same places). Setting up a variable (humidity) with a control lets learners reason about conditions that cause guttation rather than memorising a definition.	Teacher observation checklist during walk and set-up: (a) uses hand lens properly; (b) records colour, texture, location, time; (c) keeps to safety rules; (d) sets up control and labels it. Check logbook entries for specific, honest observations (e.g., 'amber, hard, brittle gum at wound on trunk' rather than 'sap'). Indicators of integrity: records no droplets if none appear and gives a possible reason. Quick oral check: 'Which waste routes did you actually see today?'

	available) in the portfolio. Part B, Guttation experiment (POE): Each group waters a potted seedling (maize, bean, or tomato seedling; or grass) well in the afternoon, places a bell jar (or clear plastic bag tied loosely with a stick to avoid contact) over it to create high humidity, and sets up a control seedling uncovered in a drier spot. They also place a plastic bag over a leafy branch (transpiration check) to see water vapour droplets inside. They predict what will appear at leaf edges/tips overnight. Next morning, they use hand lenses to examine leaf margins for droplets at hydathodes, record number and position of droplets, compare with the control, and compare to droplets inside the bagged branch. Optionally, press a clean slide gently to leaf edge or lift a droplet and place it on a slide to look under the microscope for dissolved crystals once it dries.	droplets appear at leaf tips/margins of grass and arrowroot in the early morning, and that the control may show droplets too if humid. Next day, ask: 'Where precisely are the droplets: on the leaf surface or at the edges? Are they at the same spot every time?' Direct observation notes: use a checklist to record whether each group uses the hand lens correctly and records without exaggerating.		
Explain Phase  VIDEO: 04 How Diffusion Works [HD Animation] Source: Seavuria  Search ARES for similar videos  HTML: Facilitated Diffusion (Flexbook) Source: CK-12  Search ARES for similar readings	Groups share findings in a gallery of logbook pages and a class data table: Route of waste removal, where observed, what was released, evidence. They then use Claim-Evidence-Reasoning (CER) to answer: 'How do plants eliminate wastes?' Each group writes a claim about at least one route, cites evidence from the walk or experiment, and gives reasoning linking to the process: diffusion of gases (CO₂, O₂ at night, water vapour) through stomata and lenticels; transpiration losing water vapour; exudation of gums, resins and latex through cuts or bark; guttation of liquid water with dissolved salts and sugars from hydathodes when	Say: 'Make a claim, back it with evidence you saw, and explain the science that connects them.' Give 3 minutes silent thinking, then Think-Pair-Share for 3 minutes. WAIT TIME of 10-15 seconds after: 'Is a guttation droplet the same as dew? What evidence would show the difference?' Cold-call 8 learners (one per group) to share one CER, and ask 3 different learners to restate a peer's reasoning. Press for precision: 'Which part of the plant? What was lost: gas, water, or dissolved substances?' Address misconceptions from the Predict phase: 'Plants do not sweat, but guttation and transpiration both lose water; what differs?' Use	Claim-Evidence-Reasoning with Think-Pair-Share: requires learners to base explanations on their own observations and to justify them using biological principles, building scientific argumentation. The comparison of guttation and transpiration (Venn diagram) sharpens concepts that learners often confuse.	Collect CER statements and check: claim names a specific route; evidence refers to a real observation (e.g., 'droplets at leaf tips under the jar'); reasoning links to a mechanism (e.g., root pressure, humidity reducing transpiration). Indicators of mastery: can state two differences between guttation and transpiration; correctly identifies exudates as gums, resins and latex. Exit Ticket: 'Contrast guttation and transpiration in 2 sentences.' Sort responses into got it / partly / not yet for next lesson's starter.

	roots absorb water faster than transpiration removes it (high humidity, warm moist soil, cool night); abscission of old leaves, bark and fruits storing wastes. Learners then contrast guttation and transpiration in a Venn diagram and discuss the phenomenon: why are gum arabic and latex valuable? (Gum arabic for food and pharmaceutical use, adhesives and inks, Acacia senegal in Turkana, Marsabit and Garissa; latex and resins used for rubber-like products, incense, and traditional medicines; mention commercial value linked to Financial Literacy.) Learners return to the predictions and mark which were supported or changed.	guided contrast: transpiration is water vapour through stomata, mainly in the day, in dry or warm conditions; guttation is liquid water with dissolved wastes/minerals through hydathodes, mainly at night or early morning in humid conditions. Link back to the driving question: 'These routes help with gases and liquids, but how do trees clear the waste they cannot evaporate?' Connect to the community: ask learners to share names of gum, latex or resin products they know (e.g., gum arabic, aloe, rubber, Euphorbia hedges, incense myrrh and frankincense).		
Driving Question Board (DQB) Creation  VIDEO: 05 How Facilitated Diffusion Works [Hd Animation] Source: Seavuria  Search ARES for similar videos  EXERCISE: Facilitated Diffusion Practice Source: CK-12  Search ARES for similar readings	Learners revisit the class DQB. Each group writes on sticky notes: (1) one question that has been answered (with the evidence), and (2) one new question that has emerged (e.g., 'What happens to wastes that cannot evaporate or ooze out?', 'Why do some trees make gum only when cut?', 'Is the milky Euphorbia sap poisonous because it is a waste?', 'Who buys gum arabic and for how much?'). They place notes under categories: Gaseous wastes, Liquid wastes, Solid/insoluble wastes, Economic importance. The teacher highlights the unanswered question about non-gaseous or insoluble waste products and learners vote for questions to investigate next. Learners also begin their portfolio project: planning interviews with community members (parents, farmers, traders) about uses of plant products like gum, latex and resin, observing the Drugs and	Say: 'Which of our questions can we now answer with evidence? Which new questions did the walk raise?' Give 2 minutes for silent writing. Cold-call 5 learners to read a new question aloud and say why it matters. WAIT TIME 10-15 seconds before asking: 'Which of these questions is about wastes that are not gas and not liquid?' Circle the key question: 'What happens to non-gaseous or insoluble waste products that cannot evaporate or ooze out easily?' Introduce the portfolio: 'This week, interview two people about plant products and their uses; photograph the plant and its product.' Provide interview guide: What plant? What product? How is it used? Is it sold? Where and for how much? Remind about respect, permission and safety during interviews, and discuss why some plant products are dangerous if misused.	Driving Question Board check-in and Brainstorming: making the class's growing understanding and remaining gaps visible gives learners ownership of the inquiry and motivates the next investigation.	Check DQB notes: new questions should build on evidence (not just recall), refer to a route or product, and be testable. Indicators: at least half of groups link a new question to the observed phenomenon; learners can point to a note and say which evidence answers it. Peer review: groups swap sticky notes and give one 'star' (strength) and one 'wish' (improvement).

	Substance Abuse sensitisation (e.g., some plant products are misused, such as miraa or tobacco).			
Model Building Phase  VIDEO: Diffusion and osmosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Diffusion and Effusion Practice Source: CK-12  Search ARES for similar readings	Learners revise the class model of plant excretion started in Lesson 1. On a large chart (or in logbooks), each group draws a whole plant (Acacia tree or maize plant) and labels exit routes: stomata and lenticels (gas diffusion and water vapour), hydathodes (guttation droplets), cuts or bark (exudation of gum, latex and resin), and old leaves, fruits and bark (tissue fall). They add arrows for the waste type leaving, a short note on conditions (humid night, wounding), and a '?' for wastes they cannot yet account for (insoluble wastes stored in cells, vacuoles, heartwood). They add an economic note box linking gum arabic, latex and resin to Kenyan markets. Groups present their model briefly in a 2-minute gallery presentation, and peers give feedback.	Say: 'Add what we now know. Mark with a question mark what we still cannot explain.' Provide 8 minutes for drawing. Circulate and ask: 'Which evidence supports this arrow?' and 'Why did you place the question mark there?' WAIT TIME of 10-15 seconds before group presentations. Cold-call 4 groups to present, and ask 2 other learners to summarise the presenters' key idea. Distribute a peer-feedback form for the portfolio presentation plan (clarity, evidence, use of photographs). End with the exit ticket: 'Contrast guttation and transpiration in 2 sentences.' Preview Lesson 3: 'What happens to waste products that cannot evaporate or ooze out easily? We will look at where plants store them and how they are discarded.' Remind learners to bring two plant product samples or photographs for the portfolio.	Cumulative model building and Portfolio project presentation: translating evidence into a diagram forces learners to organise and connect ideas, and revealing 'gaps' in the model motivates the next lesson. Peer evaluation of portfolio plans builds Learning to Learn and accountability.	Model rubric: (1) includes at least four exit routes with correct labels; (2) matches waste type to route (gas, liquid, resinous/latex, solid tissue); (3) includes at least one condition or evidence statement; (4) identifies a gap with a question mark; (5) has an economic note. Exit ticket responses sorted into correct, partially correct and misconception. Peer evaluation of portfolio plan: one strength and one improvement per group.

D. TEACHER REFLECTION

After the lesson: Did every group record honest observations, including when guttation droplets were few or absent? Which misconceptions persisted (dew vs guttation, plants 'sweat', exudates as only 'sap')? Were safety rules around Euphorbia latex followed? Was the class of 40-50 managed effectively in groups of 5, with every learner having a role? Which groups struggled with the CER structure, and what scaffold (sentence starters) would help next time? Did weather (dry or cloudy mornings) affect the guttation results, and should a back-up demonstration (such as photos or a pre-prepared seedling) be used? How well did learners connect gum and latex to Kenyan markets? Plan to open Lesson 3 with the unanswered question about insoluble wastes, using the exit ticket data to adjust the starter.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed hardened amber gum on Acacia and milky latex oozing from a cut Euphorbia (exudation); lenticels on stems; yellow fallen leaves and leaf scars (tissue fall); droplets at the edges of grass and arrowroot leaves, and droplets forming at leaf margins of the potted seedling under the bell jar compared with the control (guttation); water vapour droplets in the plastic bag over a
----------------------------	---

	branch (transpiration).
What did I learn?	Plants eliminate gaseous and liquid wastes by diffusion through stomata and lenticels, transpiration (water vapour), exudation (gums, resins, latex), guttation (liquid droplets from hydathodes in humid conditions), and by shedding leaves, bark and fruits. Some of these products, such as gum arabic and latex, have commercial value in Kenyan markets.
How does this explain the phenomenon?	Plants have no kidneys, but their wastes leave through several surfaces and structures: gases and water vapour diffuse out through stomata and lenticels, while liquid wastes and dissolved substances leave by guttation or exudation. This explains the oozing sap and leaf-edge droplets seen locally. However, we cannot yet explain what happens to insoluble wastes that cannot evaporate or ooze out; this is the question for Lesson 3.

LESSON 3 (80 minutes (double lesson)): Where Do the Wastes Go? Shedding Old Leaves and Locking Away Crystals

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence that plants store insoluble metabolic wastes in non-vital tissues (bark, old leaves, fruits, flowers) and remove them by periodic tissue fall, by comparing young green leaves with ageing yellow/brown leaves and testing for calcium oxalate crystals under a light microscope.
Knowledge	a) Describe ways through which plants eliminate waste products (deposition and organ/tissue fall). Learners explain that insoluble wastes such as calcium oxalate crystals, tannins and resins are deposited in non-vital organs (bark, old leaves, fruits, flowers) which are later shed. b) Recognise that these wastes are stored in a form that does not harm living cells.
Skills	SEP 3 Planning and carrying out investigations (dissecting and preparing slides); SEP 4 Analysing and interpreting data (comparing young and old leaves); SEP 6 Constructing explanations using Claim-Evidence-Reasoning; SEP 2 Developing and using models (revising the class model).
Attitudes	Respect: considerate of others' opinions while brainstorming on why plants shed parts instead of having excretory organs. Integrity: honest reporting of what is actually seen under the microscope, even when no crystals are found. Self-efficacy: confidence while handling razor blades and microscopes. Learning to Learn: shares information with peers.
Key Inquiry Question	1. How do plants eliminate waste products?
Purpose in Storyline	Lessons 1 and 2 established that gaseous and liquid wastes leave through diffusion, transpiration and guttation. This lesson explains the Acacia and Euphorbia resin and the yellowing leaves that drop seasonally: solid wastes are locked away and discarded with the organ. It raises the next question: since plants make so many chemical excretory compounds (gums, latex, resins, tannins), can humans put them to economic use?
Safety Notes	Razor blades: only the teacher or trained group leaders cut, always cutting away from the body onto a tile or cardboard; sharps returned to a labelled tray. Dilute acid (e.g. 0.1 M HCl) and iodine: wear gloves and eye protection, no tasting, wash splashes with plenty of water. Euphorbia latex irritates skin and can damage eyes: do not touch it or rub the eyes; wash hands thoroughly. Handle coverslips and slides carefully. Do not collect leaves from thorny Acacia without gloves. During the field walk, keep in groups and avoid snakes, ants and bees.

B. LESSON OVERVIEW

Learners return to the phenomenon of oozing sap on cut Acacia and Euphorbia, and old yellow leaves falling. They predict where a plant keeps wastes it cannot dissolve and excrete. Groups then collect fresh green and fallen yellow/brown leaves from school trees on a short nature walk, compare them with hand lenses, and use razor blades, slides, iodine and dilute acid to examine tissue under a light microscope for calcium oxalate crystals. They use Claim-Evidence-Reasoning to explain tissue fall as excretion, add questions to the Driving Question Board, and revise the class model. A POE setup for a guttation extension is started for observation the next morning.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners look at a photo or real	Hold up the hardened gum and	Predict-Observe-Explain (POE)	Observe and note: (1) how many

 VIDEO: Van der Weyden, Deposition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Deposition Practice Source: CK-12  Search ARES for similar readings	sample of hardened Acacia gum and a yellow leaf of mango or grevillea picked from the school compound. Individually they write a prediction in their notebooks: 'Where in a plant do wastes that cannot dissolve in water end up, and what will we see if we compare a young green leaf with an old yellow leaf?' They then do Think-Pair-Share and sketch a prediction of what each leaf's cells might contain. Groups of 4-5 record a POE prediction table (Predict column) for the guttation setup: a potted seedling (e.g. maize or sukuma wiki) covered with a clear polythene bag overnight, and what they expect to see in the morning.	yellow leaf and say: 'Remember the sap oozing from the Euphorbia and the leaves falling in our compound. Plants have no kidneys. So where could the plant put wastes it cannot wash out?' WAIT TIME 10-15 seconds in silence before anyone speaks. Give 2 minutes of individual writing, 3 minutes of pair talk, then cold-call 4 pairs (not volunteers) with 'What did your partner predict, and why?' Record all predictions on the board without judging; use 'Thank you, that is a testable idea.' Ask 2 students to respectfully disagree with a peer: 'Who sees it differently, and what is your reasoning?' Pre-position POE setups across groups and say: 'Predict what you will see on the leaves by morning.'	and Think-Pair-Share. Predicting before evidence surfaces prior ideas (e.g. that plants 'sweat' all wastes out), creates cognitive conflict later, and gives learners a stake in the observations. Pair talk lets quieter learners rehearse ideas in a class of 40-50.	predictions mention a specific location (leaf, bark, fruit) rather than just 'outside'; (2) whether learners give a reason linked to prior lessons (diffusion, transpiration, guttation); (3) the number of pairs who can state a testable prediction. Record misconceptions such as 'plants have no waste' or 'all waste leaves as water' on the board for revisiting.
Observe Phase  VIDEO: Wind Erosion and Deposition Podcast Source: CK-12  Search ARES for similar videos  EXERCISE: Erosion and Deposition Practice Source: CK-12  Search ARES for similar readings	Part A, Nature Walk (15 min): Groups walk the school locality with hand lenses and collect 3 young green leaves and 3 fallen or ageing yellow/brown leaves from trees such as mango, grevillea, jacaranda, Acacia, or from grass and arrowroot (taro) leaves. They also observe bark with resin or gum (Acacia, Euphorbia) from a safe distance, and note fruits or flowers that have dropped. They record each tree's name, location and the state of the leaves. Part B, Dissect and Compare (15 min): Using hand lenses, they compare colour, texture, thickness, dryness and veins of young versus old leaves, and sketch both. Part C, Microscopy (25 min): Group leaders (supervised by the teacher) cut thin sections or scrape tissue from young and old leaves, mount them on slides in a drop of water, and view under low	Before the walk say: 'Your job is to be a detective. Bring back evidence, not opinions.' Assign roles: collector, recorder, safety monitor, photographer (if a phone is available), and rotate them. During the walk circulate and ask each group: 'What differences do you see between this leaf and that one? What might the yellow colour tell us?' Cold-call 3 groups per stop. At the microscope, demonstrate once for the class on a projected or demonstrator microscope: 'Always begin on low power. Show me you can find the tissue before you move on.' Restrict razor blade use to trained leaders and remind: 'Cut away from your fingers, blade onto the tray after use.' Ask probing questions at benches: 'What shape is it? Does it dissolve when we add acid? Is it present in both leaves?' Give WAIT TIME of 10-15 seconds	Nature Walk and Direct Field Observation combined with hands-on microscopy. Learners collect their own evidence from familiar trees, so the phenomenon (falling leaves, oozing gum) is the data source. Comparing young versus old tissue builds the idea of patterns (SEP 4) and the concept that old, non-vital organs accumulate stored wastes. Using iodine and dilute acid helps learners distinguish crystals from starch and use tests as evidence.	Direct teacher observation using a checklist: (1) selects both young and old leaves correctly; (2) focuses the microscope on low power before high power; (3) uses blades and acid safely; (4) records at least 3 differences between leaf types; (5) completes a crystal count table for both leaf types; (6) accurately describes crystal shape and the response to acid. Look for evidence in sketches such as labelled crystals or clusters in old leaf tissue. Note groups needing help with focusing or slide preparation and provide a quick peer-tutor.

	then high power. They then add a drop of iodine to a second slide (to check for starch, which stains blue-black, so as to tell it apart from crystals) and a drop of dilute acid at the edge of the coverslip of a third slide (calcium oxalate crystals do not dissolve in ethanoic acid but carbonate crystals fizz or dissolve in dilute HCl), noting what happens to any needle-shaped or star-shaped crystals. Where possible a pinch of leaf ash is also mounted and tested. Learners draw what they see and record counts of crystals in 3 fields of view for young and old leaves in a table. Part D: Groups check the POE guttation setups and leave them overnight.	after each question. For groups who see no crystals, say: 'No crystals is also data. Where else could you look, for example the leaf veins or the edge?' Remind that the Euphorbia latex is observed only from a distance. Assess practical performance with a checklist.		
Explain Phase  VIDEO: Pontormo, The Entombment of Christ Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Deposition by Streams (Flexbook) Source: CK-12  Search ARES for similar readings	Groups pool data on a class table on the board (average crystal counts for young and old leaves, observable features). Each group writes a Claim-Evidence-Reasoning (CER) statement to answer: 'How do plants get rid of insoluble wastes, and how does the yellow falling leaf relate to it?' For example: Claim: Plants store insoluble wastes such as calcium oxalate crystals in old leaves and other non-vital organs and remove them when the organ is shed. Evidence: our slides showed more crystals in old yellow leaves than in young ones; crystals did not dissolve in ethanoic acid / fizzed in dilute HCl. Reasoning: wastes are locked in a harmless solid form in tissue that the plant no longer needs, then removed by leaf fall, so no excretory organ is needed. Learners link this to the oozing gum and resin from Acacia and Euphorbia bark (deposited in bark, which is shed or sealed over) and	Ask: 'What did you notice about the number of crystals in old versus young leaves?' Cold-call 5 different groups for data and record on the board. Then ask: 'Is this a pattern or a coincidence? How do we know?' WAIT TIME 10-15 seconds. Model a CER statement briefly using a different example (e.g. tannins in tea leaves from Kericho). Circulate as groups write and push on reasoning: 'Which part of your evidence supports your claim? Why does it make sense for the plant to store waste in an old leaf and not a young one?' Select 2 groups with different claims to present and ask the class to respond with 'I agree because...' or 'I would challenge...' to uphold Respect. Address misconceptions: 'Is leaf fall only the plant losing a part, or is it also removing waste?' Clarify that not all leaf fall is excretion, but it contributes to it. Correct that crystals in all cells are not harmful	Claim-Evidence-Reasoning (CER) with Brainstorming and class discussion. CER requires learners to connect their microscope data to the idea of deposition and tissue fall, ensuring explanations are evidence-based (SEP 6). Peer challenge builds argumentation and respect for different views, while linking to the phenomenon of the falling leaves and oozing gum gives the explanation real meaning.	Exit ticket (5 min, individual): 'Contrast guttation and transpiration in 2 sentences, and state one reason why shedding old leaves counts as excretion.' Success criteria: names water vapour loss via stomata for transpiration versus liquid droplets from hydathodes for guttation; and states that wastes deposited in old leaves are removed when they fall. Quickly sort tickets into 3 piles (secure, partly secure, not yet) to plan the next lesson's starter. CER quality is judged on whether the claim is supported by specific data from slides.

	to shed flowers and fruits. They also compare with guttation and transpiration from earlier lessons, and complete the Observe and Explain columns of the POE sheet after reading the next morning's results or a teacher-provided description.	because they are insoluble and stored in vacuoles or specific cells. Distribute exit tickets.		
Driving Question Board (DQB) Creation  VIDEO: Van der Weyden, Deposition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Deposition by Streams Practice Source: CK-12  Search ARES for similar readings	Learners revisit the Driving Question Board (How do trees and crops clear out their metabolic wastes without organs, and why are these wastes worth so much money?). In groups they write new questions on sticky notes or paper strips from today's work, such as 'Why are the crystals only in old leaves?', 'Do all plants shed leaves to remove waste?', 'What is in the sap of Euphorbia and why is it poisonous?', 'Can the gum from Acacia be sold and who buys it?'. They place notes on the board, group them under headings (Where wastes are stored, How wastes leave, Uses and value), and mark questions answered so far with a tick. A learner representative reports on one answered question and one new question.	Say: 'Let's check our board. Which questions can we now answer with evidence from today?' Invite 4 students (cold-call) to move notes to an 'Answered' area and justify with evidence. Then say: 'Write at least one new question this lesson raised for you.' WAIT TIME 10-15 seconds before sharing. Highlight the thread: 'We now know plants make many chemical wastes: crystals, gums, latex, resins. This leads us to ask whether people can use these products. Which notes on our board ask about that?' Point to the market questions and flag them as the focus for Lesson 4. Avoid answering questions directly; return them with 'What evidence might help us find out?'	Driving Question Board check-in. Visibly tracking questions makes learner curiosity the driver of the unit, shows progress from evidence, and connects today's discovery to the next investigation on economic importance.	Check-in on the DQB: count how many groups add at least one question that is new and evidence-linked, and how many can move a question to 'Answered' with a data-based justification. Look for questions moving from 'what' to 'why/how' and for questions on economic use, which show the storyline is being followed. Note quiet learners' contributions.
Model Building Phase  VIDEO: Van der Weyden, Deposition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Deposition by Groundwater (Flexbook)	Learners revise the class model of plant excretion (a large drawing of a tree, a leaf, and a crop such as maize or sukuma wiki). They add arrows and labels for deposition of insoluble wastes (calcium oxalate crystals, tannins, resins) in bark, old leaves, fruits and flowers, and for tissue fall (leaf, flower, fruit and bark shedding), alongside the previously included gaseous exchange (diffusion), transpiration and guttation. In small groups they update their own notebook models, including a key, and write	Display the class model and ask: 'What should we add to our model based on today's evidence?' Cold-call 5 students to each add one element and explain it. Ask: 'Where on the model does the Euphorbia sap fit? What about the yellow leaf that falls?' WAIT TIME 10-15 seconds. Challenge: 'Why can a plant tolerate crystals in old leaves but not in growing tips?' Encourage use of evidence. Have a group share their notebook model and invite peers to give feedback with 'one star and one	Cumulative model revision with Portfolio project launch. Updating a visible model makes learners integrate new evidence (deposition and tissue fall) with earlier ideas, showing how mechanisms work together (SEP 2). The portfolio task connects the science to the community and to Financial Literacy, setting up the economic importance in Lesson 4.	Check models for: (1) deposition of insoluble wastes shown in non-vital organs; (2) tissue fall shown as removal route; (3) linked to earlier mechanisms; (4) a valid explanation for the lack of excretory organs. Peer evaluation of portfolios will be used in later lessons; today, check that groups have a clear interview plan with at least 3 questions and an identified plant product. Review the summary table entries for accuracy and evidence use.

Source: CK-12  Search ARES for similar readings	a 2-sentence explanation of why plants need no kidneys. They then add a 'What we still need to know' box pointing towards economic importance (gum arabic, latex, tannins, resins). They record in the summary table what they observed, learned and explained. Finally, groups plan the interviews for the portfolio project on local uses of plant products, and are reminded to check the guttation setup in the morning.	wish'. Introduce the portfolio project: 'Ask family, neighbours or traders at the local market in Kisumu, Nairobi or Kericho how Acacia gum, aloe, tea, pyrethrum or latex are used. Photograph the plant and its product, and do not collect unsafe plants. Discuss responsibly any substance misuse risks with plant products such as khat (miraa).' Assign the portfolio and set up peer evaluation criteria for the next lesson: clarity, accuracy, evidence and photos. Close with: 'Since plants make all these chemicals, can people use them for money? That is our next investigation.'		
---	--	--	--	--

D. TEACHER REFLECTION

Which groups could explain, using their own slide data, why old leaves contain more crystals? Did any learners confuse starch with crystals, or think that all leaf fall is waste removal? Were razor blade and acid safety routines followed, and did every learner get time on the microscope in a class of 40-50? What changes (more microscopes, pre-made slides, rotating stations) would improve access? Which DQB questions show readiness to move to economic uses, and which need more evidence first?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We collected young green and old yellow/brown leaves from school trees and compared them with hand lenses. Under the light microscope we found [describe crystal shape, how many in young versus old leaves], and recorded how crystals reacted to iodine and dilute acid. We also noted resin or gum on the bark of Acacia and Euphorbia and set up the guttation experiment.
What did I learn?	Plants deposit insoluble metabolic wastes such as calcium oxalate crystals, tannins and resins in non-vital organs (bark, old leaves, fruits and flowers). These parts are shed periodically through tissue fall. Plants therefore do not need kidneys because wastes are stored in harmless forms.
How does this explain the phenomenon?	Claim: Plants excrete insoluble wastes by depositing them in old or non-vital tissues that are then shed. Evidence: old yellow leaves had more crystals than young leaves, and the crystals did not dissolve in ethanoic acid (or reacted in the specific way recorded). Reasoning: locking wastes away in tissue the plant no longer needs, then dropping it, removes the wastes without an excretory system, which also explains the gum and resin on cut Acacia and Euphorbia and the falling leaves.

LESSON 4 (80 minutes (double lesson)): Cash from Waste: Categorising Plant Excretory Products and Their Value in Kenya

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners search reference materials and digital resources to categorise plant excretory products by type, plant source and economic application, and use the evidence to explain why substances plants discard are worth money in Kenyan markets.
Knowledge	By the end of the lesson the learner should be able to outline the economic importance of plant excretory products (KICD SLO c) and recognise the importance of excretory products in plants (SLO d): latex (rubber, papain), tannins (leather tanning, tea), alkaloids (quinine, caffeine, nicotine), gums and resins (gum arabic, frankincense, myrrh), and essential oils (eucalyptus, lemongrass).
Skills	Searching for information and classifying data in a table (Obtaining, Evaluating and Communicating Information); constructing a Claim-Evidence-Reasoning (CER) argument; using a hand lens to observe latex and gum samples; presenting findings to peers.
Attitudes	Learning to Learn: shares information with others while discussing plant excretory products and their economic importance. Respect: considers others' opinions in group discussions. Integrity: reports findings honestly and credits sources. Financial Literacy: appreciates the commercial value of plant excretory products. Drugs and Substance Abuse (PCI): becomes aware that some plant alkaloids (caffeine, nicotine, miraa/khat) can be abused.
Key Inquiry Question	1. How do plants eliminate waste products? 2. How do humans use plant excretory products?
Purpose in Storyline	Lessons 1-3 established that plants lack excretory organs and eliminate wastes by diffusion, transpiration, guttation, exudation and leaf fall. This lesson answers the second half of the driving question: why the sticky sap from Acacia and Euphorbia is worth money. It shifts from HOW plants get rid of wastes to WHAT the wastes are and WHY they have value, and raises the question for Lesson 5 of how community members use these products and whether any are harmful.
Safety Notes	Euphorbia latex is a skin and eye irritant and can cause blindness: the teacher handles it only, with gloves, and learners observe dried samples or photographs; never touch the eyes after handling any sap. Wash hands with soap after handling any plant material. Learners must not taste any sample. Supervise the school-grounds walk, and keep learners away from thorny Acacia branches and snakes in long grass. Treat the discussion of substance abuse (miraa, tobacco, excess caffeine) sensitively and without stigma.

B. LESSON OVERVIEW

Learners begin by predicting which plant excretory products are valuable and why. In groups of 5 (8-10 clusters), they then gather evidence from biology textbooks, reference materials, any available digital resources and a short walk to the Acacia and Euphorbia trees, plus dried samples of gum arabic, rubber, tea leaves, quinine information and papaya latex. They categorise products by type, source and economic use in a class table, then explain in a CER argument why wastes have commercial value in Kenya, from Kericho tea and Mumias-region sugar cane belts to Coast cashew and the Rift Valley and Turkana gum collectors. The Driving Question Board (DQB) is revisited and the class model of plant excretion is extended to show that excretory products accumulate in tissues and sap and are not just discarded. The lesson closes with the question: How do local community members make practical use of these products, and are any harmful?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
-------	--------------------	---------------	----------------------	-------------------------------

Predict Phase  VIDEO: Oldenburg, Floor Cake Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: excretion plants latex tannins	Learners recall the oozing sap from the Acacia and Euphorbia trees near school. Individually they write a prediction in their notebooks: 'Which plant wastes do you think people in Kenya buy and sell, and what do you think they are used for?' They list at least three products or plants they know (e.g. rubber, tea, gum arabic, quinine for malaria, papaya). Then, using Think-Pair-Share, they compare predictions with a partner and sort them into 'sure' and 'not sure' products.	Display a sample or picture of dried gum arabic, a rubber band, a tea bag and a papaya. Say: 'All of these come from substances that a plant produces and gets rid of or stores away. Which one do you think is worth the most money, and why?' Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds before pairs talk. Cold-call 6 pairs (not volunteers) to share one 'not sure' product. Record predictions on the board in two columns, 'Products we know' and 'Products we are unsure of', without evaluating them. Say: 'Hold on to these ideas, we will test them with evidence today.'	Predict-Observe-Explain (POE) and Think-Pair-Share. Predicting first surfaces prior knowledge about tea, rubber and quinine so learners can later see where evidence confirms or changes their thinking. The pair talk gives every learner a voice in large classes of 40-50 and links the phenomenon (sticky sap) to economic value.	Observe whether learners name at least 2 products tied to a plant source (e.g. tea from Kericho, gum from Acacia). Indicators: predictions include a use or reason for value; learners distinguish known from unsure items. Note common misconceptions (e.g. 'latex is blood of the plant', 'only medicines are valuable') on a clipboard for the Explain phase.
Observe Phase  VIDEO: Hesse, Untitled (Rope Piece) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: excretion plants latex tannins	Part A (15 min, field observation): Under supervision, groups walk to the Acacia and Euphorbia trees and nearby plants (grass, arrowroot, eucalyptus, neem or lemongrass if available). With hand lenses they observe hardened gum and resin on bark from a distance (teacher shows fresh Euphorbia latex only with gloves), crush a lemongrass or eucalyptus leaf to smell the essential oil, and note drops of liquid, colour, stickiness and where the product is found on the plant. Part B (25 min, research): Back in class, each group receives a research card (latex; tannins; alkaloids; gums and resins; essential oils) and uses biology textbooks, reference materials and digital resources, where available through the teacher's device, to complete a table: Excretory product Type Plant source Kenyan economic or medicinal use. Required examples include quinine (Cinchona), caffeine (tea in	Before the walk say: 'You are scientists. Record what you see, smell and touch, but touch nothing from the Euphorbia. Latex can harm your eyes.' Circulate with a checklist and ask: 'Where on the plant is this substance found? Is it inside the plant or has it come out?' During research hand out cards in a jigsaw so every cluster covers a different product, and say: 'Find at least two plant sources and two uses and write where you found them.' Ask each group: 'Which Kenyan town or region is linked to this product?' Give WAIT TIME of 10-15 seconds after each question. Pause the class at 15 minutes and cold-call 5 groups to read out one fact and its source. Support weaker readers by pairing them with strong readers and providing a simplified fact sheet where internet access is limited.	Nature Walk and Direct Field Observation combined with guided research and classification. Seeing real gum, resin and oils anchors abstract text in the phenomenon, while the categorising table forces learners to look for patterns across types, sources and uses. The jigsaw ensures every group becomes an expert and shares information with others (Learning to Learn).	Direct teacher observation during the field walk and research using a checklist: (1) learner uses the hand lens correctly; (2) records at least 3 observations; (3) completes at least 4 table rows with correct type-source-use matches; (4) cites the source; (5) wears or avoids contact as instructed. Indicator for Integrity: groups write 'not found' instead of guessing. Collect one table row from each group for later checking.

	Kericho, coffee in Nyeri and Kiambu), rubber (Hevea latex), papain (papaya latex, meat tenderiser), gum arabic (Acacia senegal), tannins (Acacia/wattle bark for leather tanning, tea) and essential oils (eucalyptus, lemongrass). Groups also record one question and the source of each fact.			
Explain Phase  VIDEO: Oldenburg, Floor Cake Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: excretion plants latex tannins	Groups share their completed rows to build a class master table on the board or on chart paper. Learners then write a Claim-Evidence-Reasoning (CER) statement in response to: 'Why are plant excretory products valuable in Kenyan markets?' Claim: a one-sentence answer. Evidence: at least two specific products from the table (e.g. quinine, rubber, gum arabic, tea). Reasoning: link to the idea that plants accumulate waste products in tissues, latex vessels, bark and vacuoles, and humans use their chemical properties (antimalarial, sticky and sealing, tanning, flavouring). Learners also discuss financial literacy: who earns from gum arabic collection in Turkana and Marsabit and from tea in Kericho? They then compare their predictions with the evidence and name one idea that changed.	Say: 'Each group, send one speaker to add your rows to the class table.' Ask: 'What patterns do you notice? Which type of product has the most uses?' WAIT TIME 10-15 seconds, then cold-call 4 learners. Provide the CER frame: 'I claim ... because ... which shows ...'. Probe: 'Is a waste product still waste if people earn income from it? Defend your view.' Correct misconceptions gently: 'Latex is not blood, it is a milky fluid stored in special cells.' Add the key link: 'Many of these products are stored rather than expelled, which is why plants accumulate them in bark, latex and old leaves.' Name the PCI: 'Caffeine and nicotine are alkaloids. Why might too much of them harm a person?' Invite respectful disagreement with: 'Add on, challenge or build on a classmate's idea.' Cold-call 5 learners to read CERs aloud.	Claim-Evidence-Reasoning (CER) synthesis supported by brainstorming discussion. CER requires learners to move from a list of facts to an argument that explains why wastes have value, linking evidence to the concept of excretory products and the economy. Comparing with their predictions makes thinking changes visible.	Daily Exit Ticket: 'Contrast guttation and transpiration in 2 sentences, and name one plant excretory product with its use.' Indicators of mastery: CER includes a clear claim, at least two accurate products matched with sources, and reasoning that links accumulation or release of waste to human use. Look for correct classification of quinine and caffeine as alkaloids and rubber and papain as latex products. Sort exit tickets into 'secure', 'partly secure' and 'not yet' piles to plan Lesson 5.
Driving Question Board (DQB) Creation  VIDEO: Oldenburg, Floor Cake Source: Khan Academy (English - US curriculum)  Search ARES	Groups revisit the Driving Question Board. They place a sticky note answering part of the driving question (for example, 'Quinine from Cinchona bark treats malaria, so a waste is worth money') on the board next to existing questions. They also add new questions, such as 'How do people in our village use these products?', 'Is	Say: 'Look at our board. Which questions can we now answer with evidence from today?' Give each group 2 minutes to choose one question to move to 'answered' and one new question to add. WAIT TIME 10-15 seconds before calling on groups. Cold-call 5 groups to read their sticky note and justify the move with evidence.	DQB check-in and Brainstorming. Returning to the board shows learners that science progresses through questions, ties today's evidence to the driving question, and generates the purpose for Lesson 5. Brainstorming new questions gives learners ownership of the inquiry.	Check-ins on the DQB: indicators are that each group correctly moves at least one question with a justification using today's evidence and contributes at least one new, testable question about community use or harm. Note whether questions refer to specific products or plants, signalling deeper thinking.

for similar videos READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion plants latex tannins	rubber latex the same as Euphorbia latex?' and 'Are any of these products dangerous?' Learners mark which earlier questions are now answered, partly answered or still open.	Highlight the new question on community use and harm: 'This will lead us to our next investigation, where you will talk to people in the community.' Introduce the project: learners will interview community members about plant products and compile a portfolio of photographs. Remind them to follow ethics: ask permission, be respectful and be honest in recording responses.		
Model Building Phase VIDEO: Oldenburg, Floor Cake Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion plants latex tannins	Learners revise the class model of how plants eliminate wastes. Using a drawing of a plant (Acacia tree with arrowroot and grass below), they add labelled arrows and annotations showing where excretory products end up: latex in latex vessels and bark, gums and resins from stem wounds, tannins in bark and leaves, alkaloids in roots, bark and leaves, essential oils in leaf glands, plus guttation and leaf fall from Lessons 1-3. They add a new section called 'Value to people' linking each product to a Kenyan use. In groups they swap models with another group for peer evaluation using a simple rubric (accurate labels, correct products, Kenyan use shown, clear drawing) and give one 'glow' and one 'grow'.	Say: 'Update the model we have been building since Lesson 1. What new parts must we add so it explains why plant wastes are valuable?' Distribute the rubric and chart paper. Circulate and ask: 'Where is this product stored? What would happen if the plant did not get rid of it?' WAIT TIME 10-15 seconds. Cold-call 4 groups to explain one new element in their model. Facilitate peer evaluation with the sentence stems 'One thing that works well is ...' and 'One thing to improve is ...'. Conclude by revisiting the driving question: 'We now know how some wastes are removed and why some are worth money. Next lesson we find out how communities use them and whether any can harm us.' Distribute the interview guide for the portfolio project as take-home work.	Cumulative Model Building with Portfolio peer evaluation. Revising the model integrates mechanisms of elimination (Lessons 1-3) with the economic significance of products (Lesson 4), showing the full explanation of the phenomenon. Peer evaluation sharpens scientific communication and honesty.	Peer evaluation of research portfolios and models using the rubric. Indicators: model shows at least four product types with correct plant parts and Kenyan uses; arrows and labels are accurate; learners give specific, respectful feedback. Teacher observation notes which groups still confuse excretion with secretion or storage so this can be followed up in Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on: (1) Did learners connect the sticky sap on Acacia and Euphorbia to the commercial products they researched? (2) Which products (quinine, rubber, papain, gums, tannins) were best understood and which caused confusion, for example between latex, gum and resin? (3) Was access to digital resources adequate, or should more printed fact sheets be prepared? (4) Did the cold-calling and WAIT TIME draw out quieter learners in large clusters? (5) Were the safety rules around Euphorbia latex followed and did any learner touch their eyes? (6) Did the discussion of caffeine, nicotine and miraa remain respectful, and what do learners already know that can feed into Lesson 5's focus on community use and harm? Use exit-ticket results to decide who needs reteaching and plan interview logistics for the portfolio project.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed hardened gum and resin on Acacia bark, Euphorbia latex (teacher-handled), and the smell of essential oils in crushed lemongrass or eucalyptus leaves, and we researched products such as rubber, quinine, papain, tannins, caffeine and gum arabic.
What did I learn?	We learned that plant excretory products fall into types: latex, tannins, alkaloids, gums and resins, and essential oils. Each comes from particular plants and has uses in medicine, industry and food, and many are sold in Kenyan markets, for example tea from Kericho, gum arabic from northern Kenya, and quinine for malaria.
How does this explain the phenomenon?	Plants lack kidneys, but they store or release waste chemicals in latex vessels, bark, leaves and sap. Humans use the chemical properties of these substances, so what is waste to the plant is valuable to people, which answers why Acacia and Euphorbia products are worth money and leads us to ask how communities use them and whether any are harmful.

LESSON 5 (120 minutes (double lesson of 2 x 60 minutes; interviews and poster preparation done as homework in the preceding week)): Who Profits from Plant Wastes? Community Uses, Alkaloid Risks and the Final Explanation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate the evidence from Lessons 1-4 into a Final Explanation of how plants eliminate wastes and why these products are valuable, by presenting community interview findings on the economic importance of plant excretory products, recognising the health risks of some alkaloids, and comparing the Lesson 1 model with the final model.
Knowledge	By the end of the lesson the learner should be able to: (c) outline the economic importance of plant excretory products (medicine, sealing/adhesives, tanning, beverages, rubber, gums and resins); (d) recognise the importance of excretory products in plants; (a) describe ways through which plants eliminate waste products (transpiration, diffusion, deposition, organ/tissue fall, exudation and guttation) and link each to a product; explain that some alkaloids (e.g. nicotine, caffeine, morphine, cocaine, khat/miraa cathinone) are useful in small doses but harmful when abused.
Skills	Developing and using models (comparing the Lesson 1 model with the final model); Obtaining, evaluating and communicating information (community interviews, portfolio, poster presentation); Engaging in argument from evidence (CER matrix); Analysing and interpreting data (tallying interview results); Constructing explanations.
Attitudes	Learning to Learn: shares information with others discussing plant excretory products and their economic importance. Self-efficacy: gains confidence presenting and defending findings. Respect: is considerate of others' opinions and of community members' knowledge during presentations and peer review. Integrity: reports interview findings honestly and credits sources. Financial Literacy: appreciates the commercial value of plant excretory products. Drug and Substance Abuse (PCI): is sensitised about the effects of substance abuse while discussing alkaloids in the products found.
Key Inquiry Question	1. How do plants eliminate waste products? 2. How do humans use plant excretory products?
Purpose in Storyline	Lessons 1-4 established that plants lack excretory organs because their wastes are low in toxicity, are recycled (O ₂ and CO ₂ in photosynthesis and respiration), stored in dead tissues, and removed by gas diffusion, transpiration, guttation, exudation (latex, resin, gum) and organ fall. Lesson 5 answers the second half of the driving question, which is why the products are worth money in Kenyan markets, and completes the Final Explanation. It ends by raising the new question: How can sustainable harvesting of useful plant excretory products be balanced with environmental conservation?
Safety Notes	Interviews: work in pairs or groups of 3-4, only with known adults of the community (parents, grandparents, herbalists, farmers, gum or resin collectors, tea or coffee growers) with permission, and return before dark. Do not taste, smell closely or rub any sap, latex or unknown plant material. Euphorbia latex irritates the skin and can cause blindness, so handle only photographs or already-hardened samples with gloves and wash hands afterwards. No medicines or alkaloid-containing substances are to be brought to class. Discuss drug and substance abuse sensitively: do not ask learners to disclose personal or family substance use, and refer concerns to the school guidance and counselling teacher.

B. LESSON OVERVIEW

The anchoring phenomenon returns: Acacia and Euphorbia trees ooze sticky sap or resin when cut, grass and arrowroot leaves carry guttation droplets in the early humid morning, and old leaves fall. Learners have explained HOW plants remove wastes; today they explain WHY these products are valuable. In their groups of 5 (8-10 groups for 40-50 learners) they present the portfolios built from community interviews, with photographs/drawings of the plants and products, a use table, and survey

tallies on poster paper. They then complete a Claim-Evidence-Reasoning (CER) matrix that links the processes from Lessons 1-4 with the uses, and discuss alkaloid risks (drug and substance abuse). The lesson ends with the Final Explanation, a comparison of the Lesson 1 and final models, completion of the Driving Question Board (DQB), and a new question about sustainable harvesting for Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: excretion plants excretory products  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: excretion plants excretory products	Learners enter to find 6 labelled stations (gum arabic lump from Acacia, photo of hardened Euphorbia hedge, tea leaves from Kericho, coffee beans, a photo of the quinine/neem tree and a photo of miraa/khat). Individually, in 3 minutes, they write a prediction in their books: 'Which of these plant wastes do people in my community use or sell, what for, and could any of them harm people?' They then Think-Pair-Share with a partner and add a star beside any prediction they will check against their interview findings. Finally, they pull out their Lesson 1 model (the drawing of how a plant gets rid of wastes) and put it on the desk for later comparison.	Say: 'Remember the Acacia oozing sap near our school gate. Before our groups present, I want your best prediction: who uses these plant wastes, and for what?' Give 3 minutes silent writing, then WAIT TIME of 10-15 seconds before any pair discussion. Walk round and cold-call 4 learners (e.g. 'Akinyi, what do you predict about the gum arabic?'). Ask 'What makes you think that?' Record 3-4 predictions on the board under 'Our predictions' without evaluating them, including any prediction on possible harm. Say: 'Keep your predictions open; by the end of the lesson we will see which ones the evidence supports.' Ask learners to take out the Lesson 1 model and not to alter it yet.	Predict-Observe-Explain (POE) with Think-Pair-Share: committing to a prediction first exposes learners' prior ideas (for example that plant wastes are useless 'rubbish') so they can later be confronted with interview evidence of economic value. Think-Pair-Share gives every learner of the 40-50 a voice before the whole class discusses.	Observe whether at least 80% of learners write a prediction with a reason (look for named uses such as medicine, glue, tanning, drinks). Note on a checklist the learners who still say plant wastes are useless or who ignore the harm aspect, and target them with probing questions in the Observe Phase. Collect a quick thumbs-up/thumbs-down count on 'Can any plant waste product harm people?'
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: excretion plants excretory products  READING: <i>No closely matching reading in the ARES library for this</i>	Each group of 5 sets up its poster portfolio on a desk or wall and presents for 4 minutes in a gallery rotation (half of the groups present while the others circulate for 12 minutes, then they swap). The portfolio shows: (1) photographs or drawings of the plant and its excretory product (e.g. gum arabic from Acacia senegal in Turkana and Baringo, latex from Euphorbia used as a hedge or fish poison, tannins in Acacia/wattle bark used for tanning hides in Nakuru, tea and coffee from Kericho and Nyeri which contain caffeine, quinine	Say: 'You are now the experts. You have gathered data from our own community. Listen for evidence, not just opinions.' Set a visible timer. Circulate and ask each group: 'Which excretory process released this product: guttation, exudation, deposition, organ fall or diffusion?' and 'What is its price in the market?' After each rotation WAIT 10-15 seconds before taking answers and cold-call 3 learners per rotation (6 in total) to report one new use they learned. Prompt: 'Which community members did you	Portfolio presentation and gallery walk with peer evaluation: learners act as researchers and audience, pooling class-wide data from many households so patterns (most common uses: medicine, sealing/adhesives, tanning, beverages) emerge that no single group could see. Direct observation of real samples ties the products back to the phenomenon, and the structured recording sheet makes learners compare evidence rather than simply listen.	Direct teacher observation of portfolios against a checklist: photographs/drawings present, at least 5 plant products and uses documented, a named release process for each, at least one risk identified, and honest source notes. Peer evaluation forms (rated 1-4) are collected from every learner. Observe whether learners wear gloves and avoid touching the latex. Look for learners who confuse excretory products with secretions or foods and correct them on the spot.

activity.  Search ARES for readings Search terms: excretion plants excretory products	from cinchona and neem leaves for malaria and skin conditions, aloe sap, pyrethrum from Nakuru for insecticide, sisal wastes, rubber latex, resins and incense from myrrh and frankincense in Marsabit, miraa from Meru); (2) an interview table (name of community member, plant, product, process by which it is released, use, how it is prepared, price per kg or bundle); (3) a tally chart from the survey guide; and (4) the risks mentioned, especially misuse of alkaloids such as nicotine in tobacco, caffeine overuse, miraa/khat abuse and illegal drug plants. Visitors record at least 3 new uses in their notebooks and use a peer evaluation form (accuracy, evidence, photographs, clarity, rated 1-4). Then, with their hand lens, they re-examine a hardened Acacia gum lump and a dried Euphorbia latex sample (handled in gloves by one learner per group) and a guttation droplet sketch from Lesson 2 to link each product with its release process.	interview, and did they agree?' Insist on integrity: 'If you did not interview someone, do not write it.' Hand out gloves and supervise the sample handling; remind learners 'Do not touch the latex to your eyes or mouth.' Introduce the alkaloid risk discussion by asking: 'Tea and coffee are alkaloid products that make good money. Why do we also warn young people about miraa, tobacco and other drugs from plants?' Collect 2-3 responses and clarify that alkaloids are nitrogenous plant wastes or defence compounds that act on the nervous system; small controlled doses can be medicine, but misuse leads to addiction and health, school and family problems. Invite the guidance and counselling teacher's contacts to be listed on the board.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: excretion plants excretory products  READING: <i>No closely matching reading in the ARES library for this activity.</i>	Back in groups, learners build a Claim-Evidence-Reasoning (CER) matrix on a sheet of manila paper for the claim: 'Plants remove metabolic wastes without excretory organs, and these wastes have economic value but some carry health risks.' Evidence column: data from Lessons 1-4 (e.g. guttation droplets at the edges of arrowroot leaves at 6:30 am, water loss through stomata, latex and resin from cut stems, gum lumps, leaf fall in the dry season, gas exchange through stomata and lenticels) plus interview data from the portfolios (e.g. 7 of 12	Say: 'A claim is only as strong as its evidence. Show me which piece of evidence supports which part of your claim.' Model the first row of the CER matrix on the board using the guttation droplets. Give 10 minutes group work, then WAIT TIME of 10-15 seconds after each question. Ask: 'What evidence shows that this is excretion and not just normal secretion?' and 'Why does a plant not need a kidney?' Cold-call 5 groups to share one CER row, and ask a different group to challenge it: 'Do you agree? What evidence would you add?' Reinforce respect for	Claim-Evidence-Reasoning (CER) synthesis combined with Think-Pair-Share: the CER matrix forces learners to separate what they observed from what it means, so they connect the mechanisms (guttation, exudation, transpiration, diffusion, deposition and organ fall) to the economic uses and health risks. Peer challenge and rebuttal build argumentation skills and expose gaps in reasoning.	Check CER matrices for: a clear claim, at least 3 pieces of evidence from different lessons, at least 2 from interviews, and reasoning that mentions why plants lack excretory organs. Observable indicator of mastery: the learner explains each product using the correct process (e.g. latex = exudation; water droplets = guttation; water vapour = transpiration; fallen leaves = organ fall). Daily Exit Ticket (2 sentences): 'Contrast guttation and transpiration, and name one plant excretory product people use.' Sort tickets into got it / nearly / not yet

 Search ARES for readings Search terms: excretion plants excretory products	households use neem for malaria or skin problems; gum arabic sells at about KES X per kg; tannins used for tanning; caffeine in Kericho tea earns foreign exchange). Reasoning column: scientific explanation, e.g. 'Plants have low metabolic rates, produce less toxic wastes, recycle CO₂ and O₂ between photosynthesis and respiration, and store wastes in dead tissues, so they do not need kidneys; stored wastes such as tannins, resins, latex and alkaloids are chemically useful, so people harvest them.' Each group also writes a 3-line rebuttal to a counterclaim: 'Plant wastes are useless so we should not protect those plants.' Then they write a Final Explanation paragraph individually.	others' opinions: 'We disagree with ideas, not people.' Address the misconception that all alkaloids are harmful by asking 'What about caffeine in tea and quinine in malaria treatment?' and emphasise dosage and abuse. Link to financial literacy by asking: 'If a family earns from gum arabic or tea, what must they do to keep earning?' Leave the question open to lead to the next lesson. Provide sentence starters for learners who need support: 'The evidence shows... This means that... because...'		to plan reteaching.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: excretion plants excretory products  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: excretion plants excretory products	The class returns to the Driving Question Board started in Lesson 1. Each group takes the sticky notes of questions placed there and sorts them into 'Answered with evidence', 'Partly answered' and 'New questions'. A representative from each group sticks the notes on the board with a one-line answer and the lesson number where the evidence came from (e.g. 'Why do Euphorbia trees ooze? L3, exudation of latex'). Learners then write at least one new question on a sticky note, such as 'If we keep cutting Acacia trees for gum, will they die out?' or 'Who protects wild plants in Kenya?' and place it under 'Next questions'. Finally, they individually record the answer to the driving question: 'How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste	Say: 'Our board started with many questions. Let us see which ones we can now answer with evidence.' Allow 8 minutes of sorting. WAIT 10-15 seconds after reading each unanswered question aloud. Cold-call 5 learners (not volunteers) to explain one answered question and cite the evidence ('Which lesson and which observation?'). Ask: 'What question is still left that we cannot answer with what we know?' Guide the class to notice the thread of sustainability, then point to the group's questions about overharvesting, deforestation and conservation and say: 'Next lesson we will investigate how to balance harvesting and conserving.' Record the number of questions answered on the board to show progress. Keep unanswered questions visible.	DQB check-in and question sorting: revisiting learners' own questions shows how their understanding has grown and keeps the inquiry student-driven. Citing the lesson of origin makes the evidence trail explicit, and generating new questions motivates the next learning cycle.	Count the questions moved to 'Answered' with a correct cited piece of evidence; the target is at least 70% of the board. Observe whether the new questions from learners relate to sustainability, conservation or health (indicator of transfer). Check that no learner leaves the DQB task with a blank answer to the driving question.

	products worth so much money in our markets?'			
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: excretion plants excretory products  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion plants excretory products	Learners take out their Lesson 1 model. On a fresh page they draw a Final Model of a whole plant (e.g. an Acacia tree with an arrowroot plant beside it) that shows: wastes produced (CO ₂ , O ₂ , water vapour, tannins, resins, latex, gums, alkaloids), where they are produced or stored (leaves, stem, bark, vacuoles, dead heartwood, old leaves), the routes of removal (diffusion via stomata and lenticels, transpiration, guttation at hydathodes, exudation from cuts, deposition in tissues and organ fall), and arrows to human uses and their economic value, with a caution symbol for alkaloid misuse. Using two colours of pen they compare: blue for ideas that were in the Lesson 1 model and are still correct, and red for new or corrected ideas (for example, 'Lesson 1: plants have no wastes' becomes 'Final: plants produce wastes but remove or store them in different ways'). They then write the Final Explanation (6-8 sentences) and complete the summary table: What we observed / What we learned / How we can explain it.	Say: 'Compare your first model with what you now know. What did you get right, and what did you change?' Give 15 minutes for drawing and 5 minutes for writing. Circulate with a checklist of required model components, asking: 'Where is the waste stored? How does it leave? Who uses it?' Cold-call 4 learners to describe one change between the Lesson 1 and final model, after WAIT TIME of 10-15 seconds. Ask pairs to swap models and give one 'star' (something strong) and one 'wish' (something missing). Close with the exit ticket and say: 'Our products are valuable, but trees do not grow back overnight. Think about how to harvest without harming the plants. That is Lesson 6.' Collect the Final Explanations and the exit tickets for marking.	Model comparison (Lesson 1 vs final model) and Final Explanation writing: seeing their own corrections in a different colour makes conceptual change visible, and the written explanation links the mechanisms, products, uses and risks into one coherent account of the anchoring phenomenon.	Use a model rubric: (1) at least 4 waste products named, (2) at least 4 removal routes correctly shown (diffusion, transpiration, guttation, exudation, deposition, organ fall), (3) at least 2 economic uses, (4) an alkaloid risk noted, (5) a reason why plants lack excretory organs, (6) at least 3 visible changes from the Lesson 1 model. Peer star-and-wish feedback is reviewed. Learners scoring below 4 of 6 receive a short reteach with the CER matrix in the next lesson.

D. TEACHER REFLECTION

1. Did all groups present honest and evidence-based portfolios, and which interview data were most useful in building the CER arguments? 2. Which learners could still not connect the processes (guttation, exudation, transpiration, organ fall) to the correct product, and what scaffolds helped them? 3. How did the class respond to the discussion on alkaloids and substance abuse, and were any learners referred to the guidance and counselling teacher? 4. With 40-50 learners, did the gallery rotation give every learner a voice and enough time? 5. How many DQB questions were answered with evidence, and what do the new questions on sustainable harvesting say about learners' readiness for Lesson 6? 6. What would I change about the survey guide or timing next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What did we observe and record? (e.g. oozing gum from Acacia and latex from Euphorbia, guttation droplets on arrowroot and grass leaves, fallen old leaves, and the uses, prices and risks of plant products reported by community members in our interviews and portfolios)
What did I learn?	What did we learn? (e.g. plants lack excretory organs because they produce few toxic wastes, recycle gases, and store wastes in dead tissues; wastes leave by diffusion, transpiration, guttation, exudation and organ fall; products such as gum arabic, latex, resins, tannins, quinine, caffeine and essential oils are used for medicine, sealing, tanning and drinks and bring income; some alkaloids are harmful when abused)
How does this explain the phenomenon?	How can we explain the phenomenon in our own words? (Write a Claim-Evidence-Reasoning explanation of how trees and crops clear their wastes without kidneys and why these wastes are valuable in Kenyan markets, and state the new question: How can sustainable harvesting of useful plant excretory products be balanced with environmental conservation?)

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.